

Evolution of Image Generative Models and Their Loss Functions

Model Category	Model Name	Core Architecture	Defining Loss Function(s)
I. Variational	Variational Autoencoder (VAE)	Encoder/Decoder (U-Net style internally)	Evidence Lower Bound (ELBO) Loss: 1. Reconstruction Loss (MSE/BCE) (Fidelity) + 2. KL Divergence (Regularization/Structure)
II. Adversarial (GANs)	Standard GAN (Vanilla)	Generator (G) and Discriminator (D)	Minimax Game (Binary Cross-Entropy): D maximizes $\log D(x) + \log(1 - D(G(z)))$. G minimizes $\log(1 - D(G(z)))$.
	DCGAN	GAN using Convolutional Layers	Same as Standard GAN, but uses Spectral Normalization for stability (not a change to the loss function itself).
	Wasserstein GAN (WGAN)	GAN using a "Critic" (C)	Earth Mover's Distance (Wasserstein Loss). C minimizes the distance between P_{real} and P_{fake} , enforcing a Lipschitz constraint (initially via weight clipping).

	WGAN-GP	WGAN with Gradient Penalty	Wasserstein Loss (E.M.D.) + Gradient Penalty (GP) term, which is added to enforce the Lipschitz constraint more smoothly.
	CycleGAN / Pix2Pix	Conditional GANs (cGANs)	GAN Loss (Adversarial) + L1/L2 Reconstruction Loss (for pixel alignment) + Cycle-Consistency Loss (for CycleGAN only).
	StyleGAN / ProGAN	Progressive/ Style-Based GANs	WGAN-GP Loss (used for stability) + Path Length Regularization (for disentanglement in StyleGAN).
III. Likelihood-Based	Autoregressive Models (e.g., PixelRNN/CNN)	Sequence Models (RNN/CNN)	Negative Log-Likelihood (NLL) (Directly maximizing the probability of generating the correct next pixel/token).
IV. Diffusion Models	Denoising Diffusion Probabilistic Models (DDPM)	U-Net (ϵ_θ) (The noise predictor)	Mean Squared Error (MSE): The U-Net minimizes the difference between the predicted noise (ϵ_θ) and the actual noise ($\mathbf{z}$) added to the image. $\mathcal{L}_{\text{DDPM}} = \ \epsilon_\theta(\mathbf{x}_t, t) - \mathbf{z}\ ^2$
	Latent Diffusion Models (LDM)	U-Net applied in the Latent Space	MSE Loss on Noise (Same as DDPM), but applied to a compressed latent representation of the image instead of the full pixel image.